

Chemical Transport Modelling

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Figure 1 shows the boundary layer (BL) height at SMEAR II on two different dates: summer (10.07.2011) and winter (18.02.2011). The BL height was determined as the altitude where the Richardson number is near 0.25 (Figures 2–3).

During summer, the BL reaches about 600 meters in the daytime. This is because the sun remains high in the sky for extended periods, providing more energy to the surface. The increased surface heating causes warm air to rise and mix turbulently with the air above, resulting in a deeper convective BL.

In contrast, the winter BL height is significantly lower, with modeled daily maximums between 50-200 meters. Limited solar radiation and low surface heating suppress turbulence, leading to a shallow BL. Additionally, snow reflects sunlight and acts as an insulating layer, further reducing surface heating.

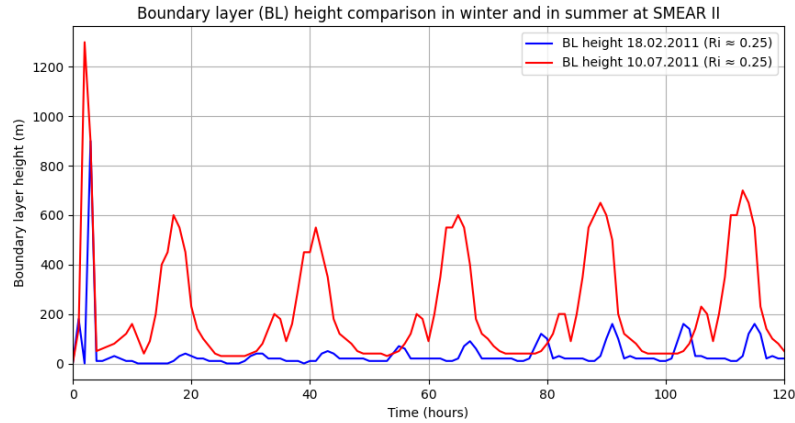


Figure 1: Boundary layer height over 5 days.

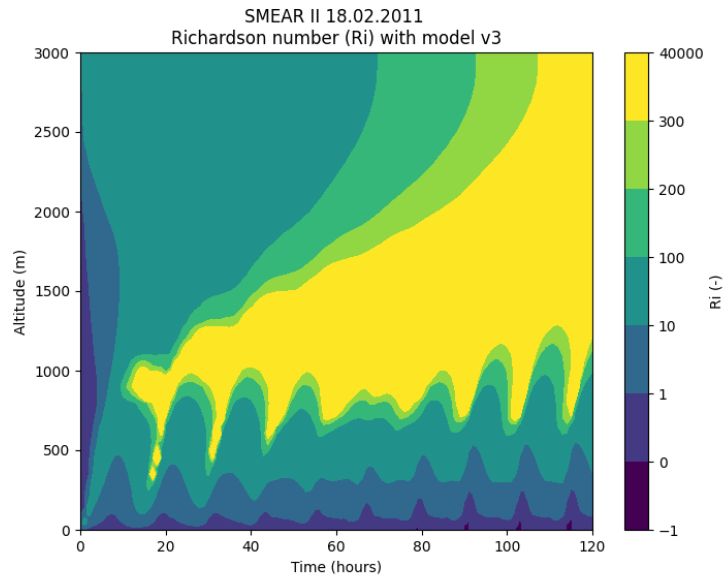


Figure 2: Richardson number during winter over 5 days.

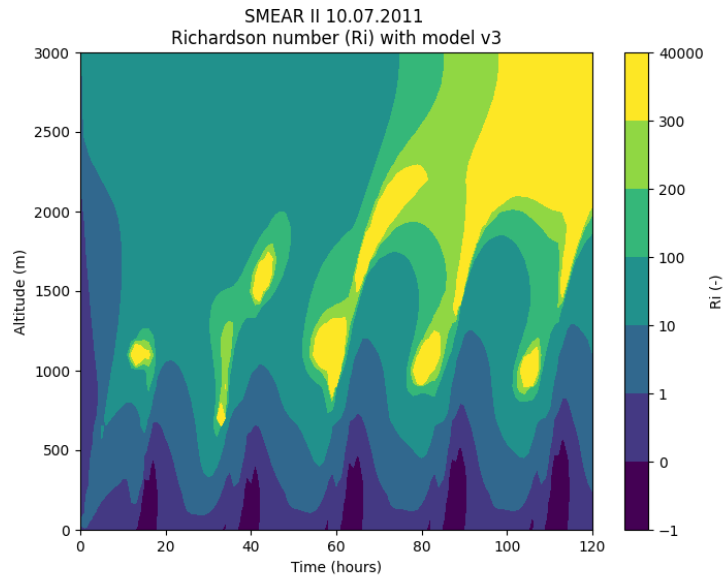


Figure 3: Richardson number during summer over 5 days.

Emissions Task

First, the relationship between isoprene and monoterpene emission rates and temperature was modelled. The results are shown in Figure 4.

The red curve (monoterpenes) shows an exponential increase with temperature. In contrast, the blue curve (isoprene) initially increases with temperature, reaches a maximum, and then decreases, reflecting an optimal temperature range for isoprene production driven by plant physiological processes.

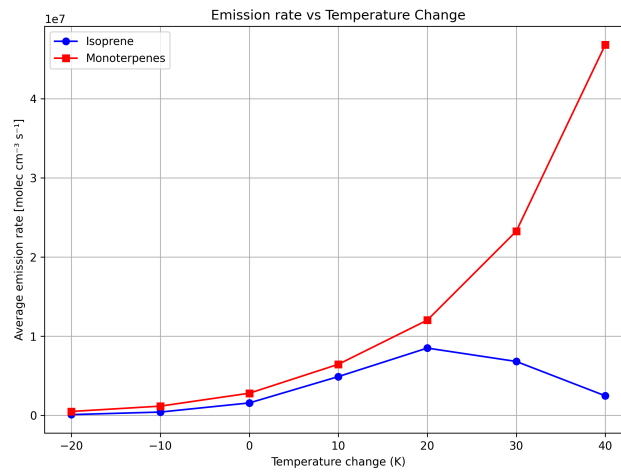


Figure 4: Isoprene and monoterpene emission rates as a function of temperature.

Secondly, the relationship between emission rates and solar radiation (photosynthetically active radiation, PAR) was modeled, as shown in Figure 5.

Here, isoprene emissions (blue curve) increase with radiation but begin to saturate at high PAR levels. Monoterpene emissions were independent of PAR. However, this assumption oversimplifies reality, since some monoterpene emissions do respond to light in certain plant species, as discussed in the lecture. This simplification is one limitation of the model as it overlooks the radiation dependence observed in real atmospheric conditions.

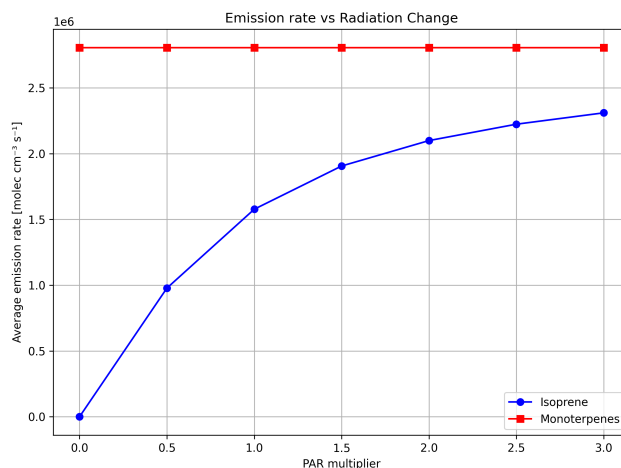


Figure 5: Isoprene and monoterpene emission rates as a function of PAR.

Chemistry Task

Figure 6 shows the concentration of H_2SO_4 as a function of the condensational sink (CS) and the SO_2 mixing ratio. Two effects can be seen: higher SO_2 mixing ratio increases the production of H_2SO_4 , while higher CS removes H_2SO_4 by condensation onto particles.

When comparing two environments: clean areas like Antarctica with low $[\text{SO}_2]$ and low CS, and polluted urban areas with high $[\text{SO}_2]$ and high CS, the H_2SO_4 concentrations are similar. This shows that having high $[\text{SO}_2]$ alone is not enough for high $[\text{H}_2\text{SO}_4]$. A low condensational sink is also needed.

$[\text{H}_2\text{SO}_4]$ is highest when $[\text{SO}_2]$ is high and CS is low. This means that both production and loss processes control the amount of gas phase H_2SO_4 in the air.

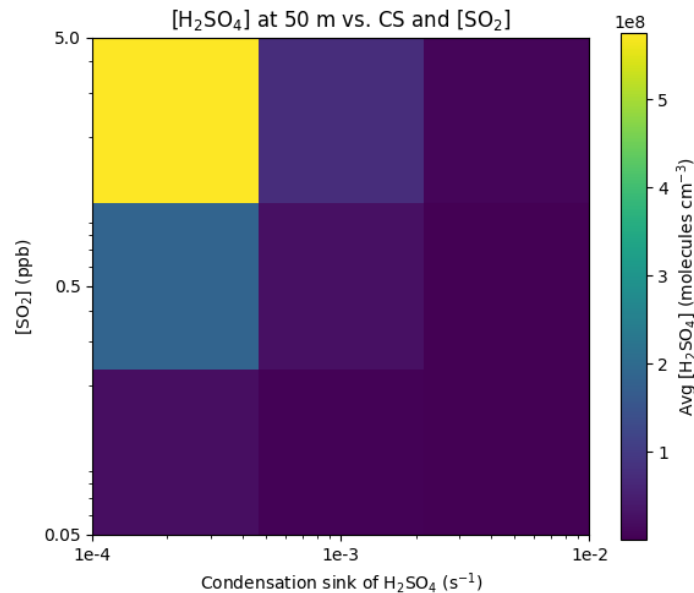


Figure 6: Modeled H_2SO_4 concentrations as a function of SO_2 mixing ratio and H_2SO_4 condensational sink.

Final Task: New Particle Formation Event at Hyltemossa

Figures 7–9 compare the modeled particle dynamics on April 6, 2018, at Hyltemossa with DMPS observations.

The modeled particle size distributions (PSD) are shown in Figures 7a and 7b. As seen in Figure 7a, the model predicts a nucleation burst and subsequent particle growth. However, the particle formation event occurs about 10 hours earlier in the model compared to observations. The zoomed-in PSD in Figure 7b resembles the observed PSD, showing a shift towards larger particle sizes over time and a gradual decrease in number concentration.

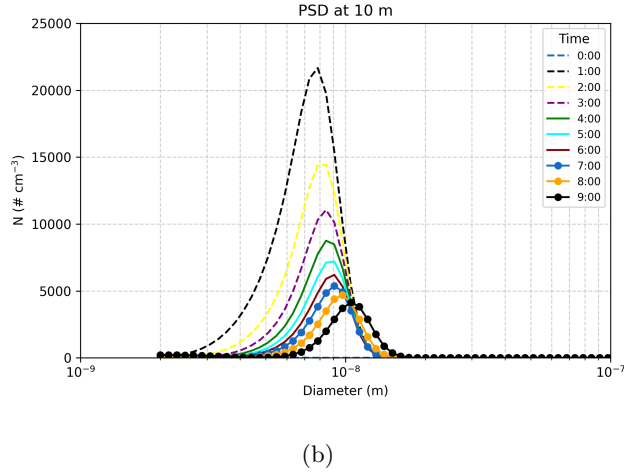
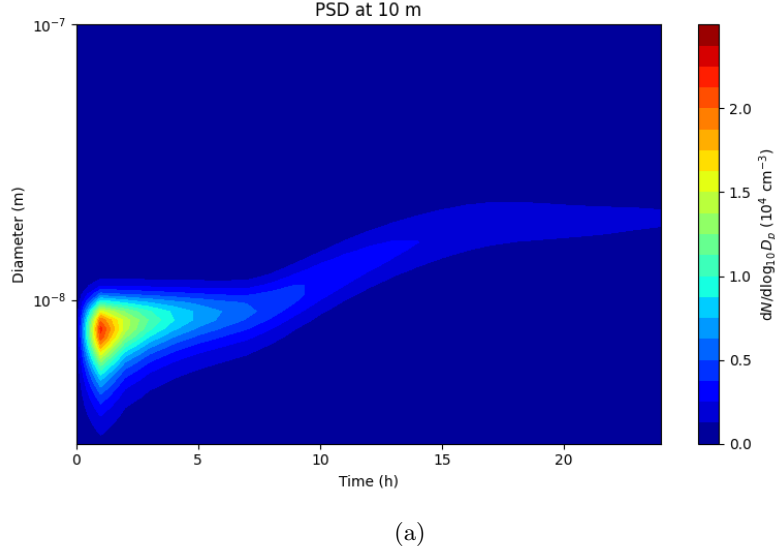


Figure 7: Particle size distribution analysis at 10 m height.

Figure 8 shows the total particle number concentration (PN) over time. Similar to the PSD, the modeled nucleation burst appears about 10 hours earlier than observed. Additionally, the modeled peak is sharper, and the subsequent decline in particle number is more abrupt compared to measurements.

Figure 9 shows the total particle volume concentration (PV). The modeled PV increases after the particle formation event, consistent with observations, but both PN and PV in the model display smoother time series compared to the experimental data.

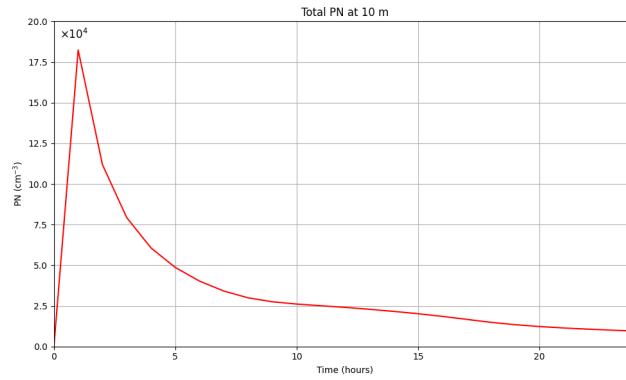


Figure 8: Total particle number concentration at 10 m height.

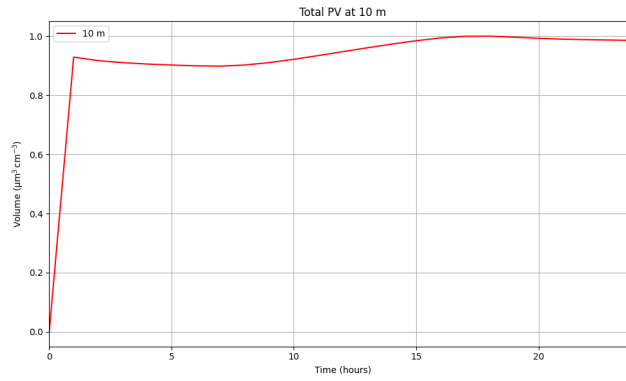
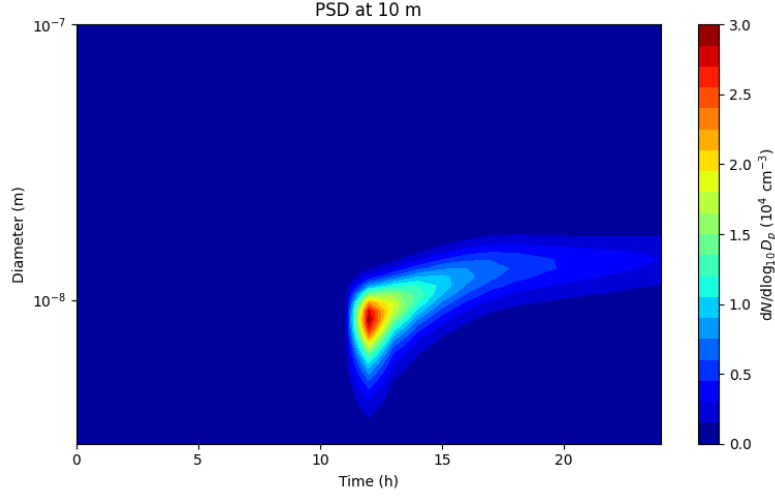
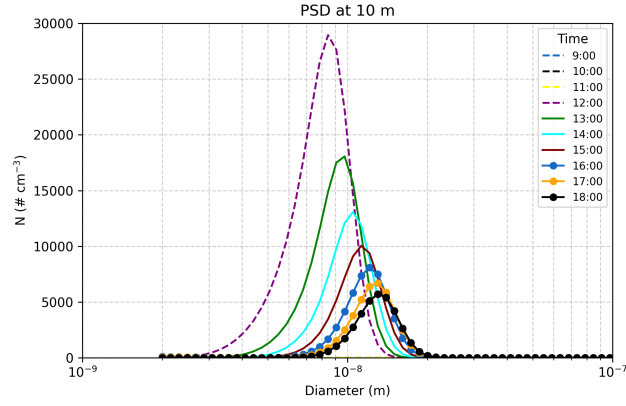


Figure 9: Total particle volume concentration at 10 m height.

Starting the aerosol processes later in the model (e.g. not the whole 5th day but in the middle of the 5th day) resulted in better alignment with the observed particle formation event timing, as shown in Figures 10a and 10b.



(a)



(b)

Figure 10: Particle size distribution analysis at 10 m height when aerosol processes start at 4.5 days.

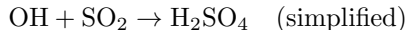
In the model, the volume growth of a single particle over a time interval Δt is given by:

$$\Delta v_p = \sum_i C_{R,i} \nu_i c_i \Delta t,$$

where $C_{R,i}$ is the collision rate between a particle of volume v_p and a gas

molecule i , and $c_{\text{eq},i}$ is the equilibrium saturation concentration at the particle surface. The other two factors — the molecular volume of species i (ν_i) and the gas-phase concentration of molecule i (c_i) — determine whether H_2SO_4 or ELVOC contribute more to the condensational growth of particles.

During the daytime, $[\text{H}_2\text{SO}_4]$ is much higher than $[\text{ELVOC}]$ because the formation of sulfuric acid



is rapid due to the photochemical production of OH radicals. Although ELVOC molecules have larger molecular volumes than sulfuric acid, the much greater $[\text{H}_2\text{SO}_4]$ means that it contributes more to particle volume growth overall. This makes H_2SO_4 the dominant driver of particle growth.

Although ELVOC formation is also enhanced by sunlight through the oxidation of BVOCs, its production rate is lower because ELVOC formation involves multiple slower reactions with lower rate coefficients.

Particle growth slows down in the evening because solar UV radiation drops, causing a reduction in OH radical production. Consequently, $[\text{H}_2\text{SO}_4]$ collapses.

BVOC emissions and their oxidation to ELVOC also wane in the evening because lower temperatures and reduced sunlight limit plant activity and photochemical reactions. With both $[\text{H}_2\text{SO}_4]$ and $[\text{ELVOC}]$ declining, the growth rate decreases and particle growth slows.